

ECON 5345 Homework 2 Report

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Question 1

a. Q1

Question 2

a. Rewriting the $ARMA(1, 1)$ process, we get

$$(1 - 0.8L)y_t = (1 + 0.4L)e_t.$$

Suppose that $y_t = \psi(L)e_t$. Then we have

$$(1 - 0.8L)(\psi_0 + \psi_1L + \cdots) = 1 + 0.4L.$$

Equating coefficients, we get the impulse responses:

$$\psi_j = \begin{cases} 1, & j = 0 \\ 1.2 \times 0.8^{j-1}, & j \geq 1 \end{cases}$$

Plotting this function, we get Figure 1. Codes in “hw2_q2.R”.

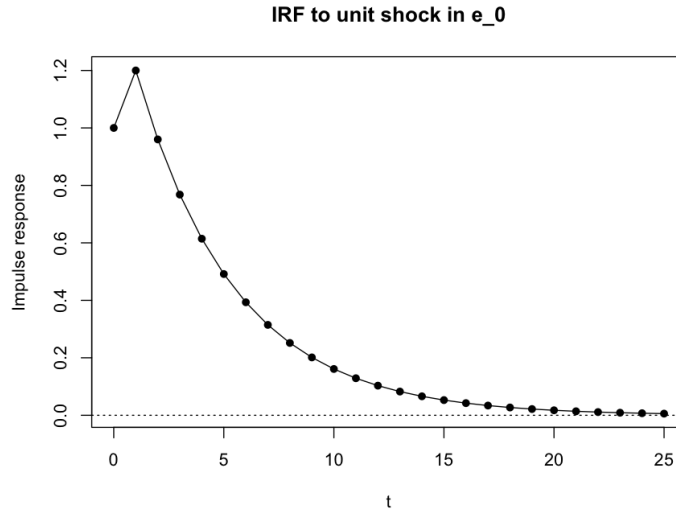


Figure 1: IRF to unit shock in e_0

b. We can write y_t as a $MA(\infty)$ process:

$$y_t = \sum_{j=0}^{\infty} \psi_j e_{t-j}.$$

It immediately follows that $\mathbb{E}[y_t] = 0$.

The variance is

$$\text{Var}[y_t] = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j^2 \right) = 3 \left(1 + \frac{1.2^2}{1 - 0.64} \right) = 15.$$

The autocovariances are

$$\gamma(1) = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+1} \right) = 3 \left(1.2 + \frac{1.2^2 \times 0.8}{1 - 0.64} \right) = 13.2.$$

$$\gamma(2) = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+2} \right) = 3 \left(0.96 + \frac{1.2^2 \times 0.8^2}{1 - 0.64} \right) = 10.56.$$

c. Since $\text{Var}[e_t] = 3$, we have the ACGF $G : \mathbb{C} \rightarrow \mathbb{C}$ defined as

$$\begin{aligned} G(z) &= 3 \left(1 + \sum_{j=1}^{\infty} 1.2 \times 0.8^{j-1} z^j \right) \left(1 + \sum_{j=1}^{\infty} 1.2 \times 0.8^{j-1} z^{-j} \right) \\ &= 3 \times \left[\left(1 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2(j-1)} \right) + \left(1.2 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2j-1} \right) z \right. \\ &\quad \left. + \left(0.96 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2j} \right) z^2 + \dots \right]. \end{aligned}$$

By the ACGF,

$$\gamma(0) = 15$$

$$\gamma(1) = 13.2$$

$$\gamma(2) = 10.56.$$

This verifies the results in part (b).

d. I used $e \sim \text{Gamma}(2, \sqrt{3})$ to generate the innovations. The figure below is the simulated path of y . Codes in “hw2_q2.R”.

The reason for burning is to gain stationarity in the long run.

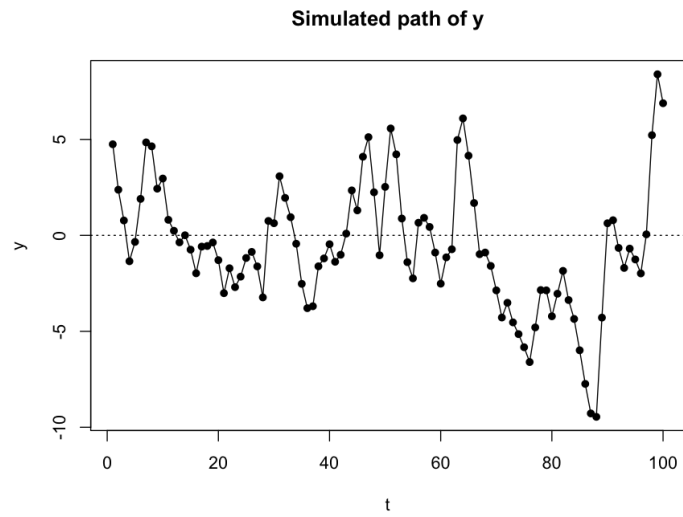


Figure 2: Simulated path of y

Question 3

a. Q3

Question 4

a. The expectation is

$$\begin{aligned}\mathbb{E}[x_t] &= \mathbb{E}[\alpha] \cos t + \mathbb{E}[\beta] \sin t \\ &= 0.\end{aligned}$$

The variance is

$$\begin{aligned}\text{Var}[x_t] &= \text{Var}[\alpha] \cos^2 t + \text{Var}[\beta] \sin^2 t + 2 \text{Cov}(\alpha, \beta) \cos t \sin t \\ &= \cos^2 t + \sin^2 t \\ &= 1.\end{aligned}$$

The second line is by the independence of α and β .

b. No. Note that

$$\begin{aligned}\mathbb{E}[\alpha^2] &= \text{Var}[\alpha] + (\mathbb{E}[\alpha])^2 \\ &= 1 \\ \mathbb{E}[\beta^2] &= \text{Var}[\beta] + (\mathbb{E}[\beta])^2 \\ &= 1 \\ \mathbb{E}[\alpha\beta] &= \text{Cov}(\alpha, \beta) + \mathbb{E}[\alpha]\mathbb{E}[\beta] \\ &= 0.\end{aligned}$$

As a result, we have

$$\begin{aligned}\mathbb{E}[x_t x_{t-k}] &= \mathbb{E}[\alpha^2] \cos t \cos(t-k) + \mathbb{E}[\beta^2] \sin t \sin(t-k) \\ &\quad + \mathbb{E}[\alpha\beta][\cos t \sin(t-k) + \sin t \cos(t-k)] \\ &= \cos t \cos(t-k) + \sin t \sin(t-k) \\ &= \cos[t - (t-k)] \\ &= \cos k.\end{aligned}$$